

A Conformal Selective Diagnosis Framework for Reliable Bearing Fault Identification Under Domain Shift

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Abstract—Deep learning models for bearing fault diagnosis provide point predictions without reliability guarantees, limiting deployment in safety-critical industrial applications. We present the first systematic evaluation of conformal prediction (CP) for bearing fault diagnosis under domain shift and class imbalance. Evaluating on the CWRU dataset with cross-domain protocols (training on loads 0–1, testing on load 3), three findings emerge: (1) CP maintains $\geq 99.9\%$ coverage under severe domain shift where classifier accuracy drops to 86.3%, outperforming MC Dropout (88.5% coverage); (2) RAPS provides the best practical tradeoff with 95.6% coverage and compact prediction sets averaging 3.82 classes; and (3) the proposed Conformal Selective Diagnosis (CSD) framework achieves 100% accuracy on automatically classified samples (4.5% of total), enabling reliable human-AI collaboration. The threshold (THR) score function is further shown to fail categorically under domain shift, motivating the mathematical explanation provided for practitioners.

Index Terms—conformal prediction, bearing fault diagnosis, domain shift, uncertainty quantification, human-AI collaboration

I. INTRODUCTION

Rolling element bearings are among the most failure-prone components in rotating machinery, accounting for 40–50% of induction motor failures [1]. In high-stakes applications such as wind turbine drivetrains and aerospace propulsion systems, a misclassified bearing fault can trigger cascading mechanical failures with severe safety and economic consequences. Consequently, quantifying the *reliability* of each diagnostic prediction—not just its predicted class—has become an industrial imperative for condition-based maintenance programs.

Deep learning (DL) achieves high accuracy in bearing fault diagnosis [2], yet produces only point predictions without rigorous probabilistic meaning [3]: a 95% softmax confidence carries no guarantee that the true fault lies in the top prediction. Uncertainty quantification (UQ) methods such as Monte Carlo (MC) Dropout [4] and deep ensembles [5] give heuristic estimates but lack formal guarantees and degrade

under distribution shift—precisely when machinery operates under novel load or speed.

Conformal prediction (CP) [6] offers a fundamentally different approach: given any classifier and error rate α , CP constructs prediction *sets* guaranteed to contain the true label with probability $\geq 1 - \alpha$, requiring only exchangeability between calibration and test samples. Unlike Bayesian or ensemble methods, this guarantee is distribution-free and holds for any underlying data distribution without modeling assumptions. Recent advances address covariate shift [7], class imbalance [8], and set efficiency [9].

Despite these advances, CP has not been systematically studied for bearing fault diagnosis. Real-world deployment poses three challenges. (1) **Domain shift**: operating conditions (load, speed, temperature) differ between training and deployment, violating the exchangeability standard CP requires; the severity of this violation—where 10–15% accuracy shifts are common—has not been quantified for CP-based diagnosis. (2) **Class imbalance**: rare fault types are underrepresented in calibration data, yet CP’s guarantee must hold across all classes, leaving open whether marginal CP suffices or class-conditional calibration is needed. (3) **Practical automation**: a set of 8 of 10 classes is not actionable; the protocol must decide which predictions proceed autonomously and which escalate to human review, balancing coverage, compactness, and throughput.

This paper presents the first systematic study of conformal prediction (CP) for reliable bearing fault diagnosis under domain shift and class imbalance. The main contributions are summarized as follows:

- 1) A **Conformal Selective Diagnosis (CSD)** framework is proposed for human-AI collaborative fault diagnosis. It calibrates prediction sets from a pretrained 1D ResNet, automatically routes singleton predictions to deployment, and defers multi-class sets to human experts for review, converting classifier uncertainty into an actionable, auditable deployment protocol.
- 2) **Practical score-function and deployment guidelines** are derived for CP-based fault diagnosis. Because probability-threshold scores collapse to singletons under domain shift (regardless of α), a principled score choice becomes essential: rank-based APS and RAPS remain robust, with APS providing conservative coverage for safety-critical settings and RAPS offering a compact and informative tradeoff (3.82 avg. set size, 56% smaller than APS) for practical deployment.

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3) A **comprehensive empirical validation** of CP is conducted under cross-load domain shift and rare-fault imbalance on the CWRU benchmark dataset. CP maintains near-perfect marginal coverage ($\geq 99.9\%$) where classifier accuracy drops to 86.3% and MC Dropout coverage falls to 88.5%, and preserves high per-class coverage for rare faults ($>98.9\%$), demonstrating that marginal CP alone eliminates the need for class-conditional calibration under realistic imbalance.

II. RELATED WORK

Deep learning for fault diagnosis has progressed rapidly. CNNs process raw vibration signals directly, bypassing hand-crafted features [2], and 1D residual networks [10] achieve state-of-the-art accuracy on time-domain benchmarks. Yet these models degrade under *domain shift*—changes in speed, load, or temperature between training and deployment. Domain adaptation [11] partially addresses this via feature alignment, but focuses on recovering point accuracy without quantifying the reliability of individual predictions.

MC Dropout [4] and deep ensembles [5] provide heuristic uncertainty estimates by aggregating multiple stochastic forward passes or independently trained models, but their link to actual error rates is purely empirical, with no formal coverage bound. Under shift, a well-documented pathology emerges: softmax outputs stay overconfident even as accuracy degrades [3], so these methods underestimate uncertainty when it matters most. Ren et al. [12] decompose aleatoric from epistemic uncertainty for fault diagnosis but likewise lack finite-sample guarantees. Table I summarizes the landscape.

TABLE I
COMPARISON OF UQ APPROACHES FOR FAULT DIAGNOSIS.

Method	Cov. Guar.	Retrain	Shift Rob.	Cost
MC Dropout [4]	No	Yes	No	High ($T \times$)
Deep Ens. [5]	No	Yes	No	Very High
Bayesian NN [13]	No	Yes	No	High
CP+APS (ours)	Yes	No	Yes	Negligible
CP+RAPS (ours)	Partial [†]	No	Yes	Negligible

[†]RAPS: 95.6% at $\alpha=0.10$. THR fails under shift.

Despite CP’s theoretical maturity and proven success in medical imaging and autonomous driving [14], its application to industrial fault diagnosis remains unexplored. The gap is significant: bearing diagnostics combines high-frequency vibration signals, real-time constraints, severe class imbalance, and routine domain shift. Existing UQ methods either provide no formal guarantees (MC Dropout, ensembles) or require per-class sample sizes unavailable for rare faults (Bayesian approaches). CP fills this gap with distribution-free, finite-sample coverage guarantees as a lightweight post-hoc wrapper requiring no retraining.

III. PRELIMINARIES

A. Conformal Prediction

Consider a K -class task with input X and label Y . Let $\hat{\pi}(x) \in \Delta^{K-1}$ denote the softmax output. Split conformal pre-

diction [6], [15] constructs a prediction set $C(x) \subseteq \{1, \dots, K\}$ satisfying:

$$P(Y \in C(X)) \geq 1 - \alpha, \quad (1)$$

for user-specified $\alpha \in (0, 1)$ (Eq. 1).

Given a calibration set $\mathcal{D}_{\text{cal}} = \{(X_i, Y_i)\}_{i=1}^n$ exchangeable with the test point, the procedure is:

- 1) Compute nonconformity scores $s_i = s(X_i, Y_i)$ for calibration samples.
- 2) Set $\hat{q} = \text{Quantile}(\{s_i\}_{i=1}^n; \frac{\lceil (1-\alpha)(n+1) \rceil}{n})$.
- 3) At test time: $C(x) = \{y : s(x, y) \leq \hat{q}\}$.

Theorem (Finite-Sample Coverage [6]). *If $(X_1, Y_1), \dots, (X_n, Y_n), (X, Y)$ are exchangeable, then:*

$$1 - \alpha \leq P(Y \in C(X)) \leq 1 - \alpha + \frac{1}{n+1}. \quad (2)$$

This guarantee is distribution-free, finite-sample, and holds for any score function.

B. Nonconformity Score Functions

We study three score functions, from simple to robust. **THR** (Threshold): $s_{\text{THR}}(x, y) = 1 - \hat{\pi}_y(x)$, the classifier’s self-reported uncertainty for class y . It is fragile: when test-time confidence drops under shift while calibration was dominated by high-confidence predictions, the threshold $\hat{q}$ is too permissive, producing singletons irrespective of true uncertainty. THR fails because it inherits the classifier’s calibration without correction.

APS [8]: Let $\sigma(x)$ order classes by decreasing predicted probability. Define $s_{\text{APS}}(x, y) = \sum_{j=1}^{k(x,y)} \hat{\pi}_{\sigma(j)}(x)$, where $k(x, y)$ is the rank of y in $\sigma(x)$. APS accumulates probability in rank order, making it robust to monotone transformations of softmax outputs.

RAPS (Regularized APS) [9]: $s_{\text{RAPS}}(x, y) = s_{\text{APS}}(x, y) + \lambda(k(x, y) - k_{\text{reg}})^+$, with $\lambda > 0$ and $(z)^+ = \max(0, z)$. The penalty raises the score of classes ranked beyond k_{reg} , trading a little coverage for much more compact sets; λ and k_{reg} are tuned on held-out calibration data without invalidating exchangeability.

C. Extensions

Weighted CP [7] addresses covariate shift by reweighting calibration scores with likelihood ratios $w = dP_{\text{test}}/dP_{\text{train}}$, but estimating these ratios in high-dimensional signal space is itself hard, limiting effectiveness under severe shift. **Class-conditional CP** [8] computes per-class thresholds $\hat{q}_y$ guaranteeing $P(Y \in C(X) \mid Y=y) \geq 1 - \alpha$, but each $\hat{q}_y$ uses only n_y samples, requiring $\Omega(1/\alpha)$ per class and underperforming marginal CP when rare classes are scarce.

IV. METHODOLOGY

The *Conformal Selective Diagnosis* (CSD) framework addresses three challenges: coverage under domain shift, fairness for rare faults, and human-AI collaboration. As shown in Fig. 1, the pipeline has three phases: (1) a 1D ResNet produces softmax probabilities $\hat{\pi}(x)$; (2) calibration yields a conformal threshold $\hat{q}$; (3) prediction sets route singletons to automation and multi-element sets to human experts.

CSD: Conformal Selective Diagnosis Framework

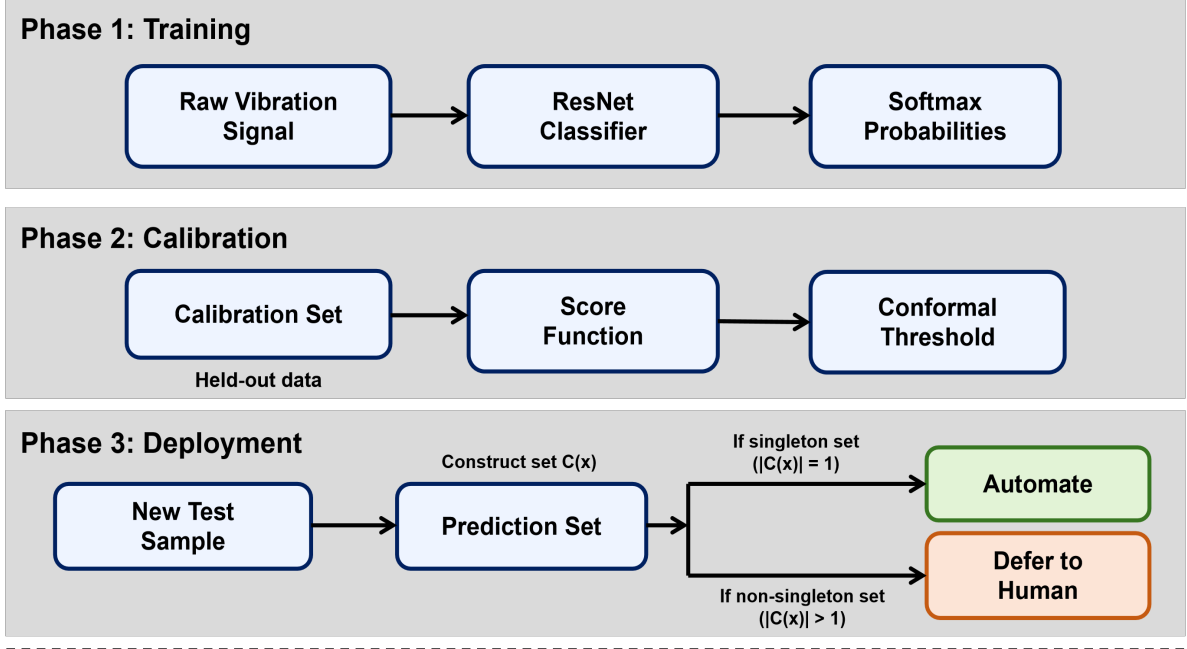


Fig. 1. CSD framework overview: Phase 1 trains a 1D ResNet for softmax probabilities; Phase 2 calibrates conformal thresholds; Phase 3 routes singletons to automation and multi-class sets to human experts.

A. Dataset and Setup

We use the CWRU bearing dataset [16], the de facto standard benchmark [17]. Drive-end accelerometers sample at 12 kHz from a 2 HP motor under four loads (0–3 HP, 1797–1730 RPM). Faults are EDM-machined at three diameters (0.007, 0.014, 0.021 in.) at the inner race (IR), ball (B), and outer race (OR), yielding $K = 10$ classes including Normal (naturally overrepresented, $\approx 4\times$ any fault class). Signals are segmented into non-overlapping $L = 256$ windows and standardized per window. The protocol uses Loads 0–1 as source ($\approx 24,700$ samples, 70/30 train/calibration) and Load 3 as target ($\approx 12,367$ test).

B. Classifier and Training

A 1D ResNet [10] ($\sim 816K$ parameters) with three residual blocks (64 $\rightarrow$ 128 $\rightarrow$ 256 filters, kernel 3), global average pooling, and a softmax head operates directly on raw signals. Training uses Adam (lr = 10^{-3} , weight decay 10^{-4}) with cosine annealing over 120 epochs at batch size 256. MC Dropout uses $p = 0.3$, $T = 50$ passes.

C. Evaluation Protocol

Three shift levels are evaluated: **In-Domain** (train/test on Loads 0–1), **Mild** (test on Load 2), and **Severe** (test on Load 3, the primary setup, $\approx 14\%$ accuracy drop from 99.9% to 86.3%). Four metrics are reported: *empirical coverage*, *average set size* ($|C(x)|$), *automation rate* (fraction of singletons), and *automated accuracy* (accuracy on singletons, which carry no coverage guarantee but are deployed autonomously).

Algorithm 1 formalizes CSD; the automation rate $r_{\text{auto}} = |\{i : |C(X_i)| = 1\}|/n$ quantifies operational efficiency.

Algorithm 1 CSD: Conformal Selective Diagnosis

Require: Classifier $\hat{\pi}$, calibration $\mathcal{D}_{\text{cal}}$, score s , level α

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1: Compute scores  $\{s(X_i, Y_i)\}_{i=1}^n$  on  $\mathcal{D}_{\text{cal}}$ 
2:  $\hat{q} \leftarrow \lceil (1-\alpha)(n+1) \rceil / n$  quantile of scores
3: for each test input  $x$  do
4:    $C(x) \leftarrow \{y : s(x, y) \leq \hat{q}\}$ 
5:   if  $|C(x)| = 1$  then
6:     Automate: output singleton
7:   else
8:     Defer: present  $C(x)$  to expert
9:   end if
10: end for

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Ensure: $P(Y \in C(X)) \geq 1 - \alpha$

V. EXPERIMENTS AND RESULTS

All experiments use the cross-domain protocol (Loads 0–1 source, Load 3 target, severe shift, 70/30 train/calibration). The 1D ResNet achieves 86.3% test accuracy—a 13.6-point drop from in-domain 99.9%. RAPS uses $\lambda=0.1$, $k_{\text{reg}}=3$; MC Dropout $T=50$, $p=0.3$.

A. Coverage Validation

Table II shows APS achieves $\geq 99.9\%$ coverage across all α , confirming the finite-sample guarantee (Eq. 2); Fig. 2 visualizes coverage, set size, and singleton rate. MC Dropout reaches only 88.5% at $\alpha=0.10$ and 94.3% even at $\alpha=0.01$ —well below target. The gap is structural, not a tuning issue: its variance is tied to weight-space geometry, not input-space

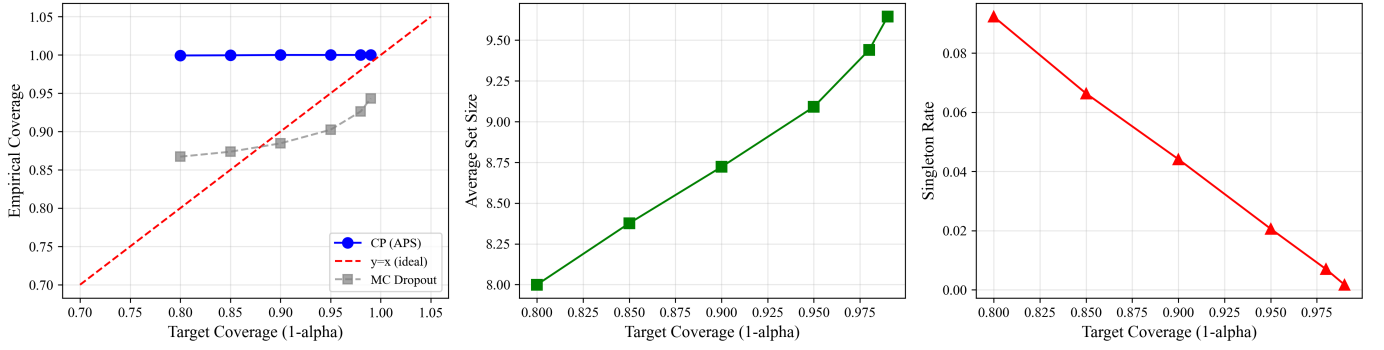


Fig. 2. Coverage validation under domain shift: coverage vs. target, set size, and singleton rate across α .

shift, so no number of forward passes recovers the guarantee. The monotonic set-size growth as α decreases (8.00 to 9.64) shows CP properly trading compactness for coverage.

TABLE II
CP COVERAGE UNDER DOMAIN SHIFT.

α	Target	APS Cov.	Size	MC Drop.
0.20	≥ 0.80	0.999	8.00	0.867
0.15	≥ 0.85	1.000	8.38	0.874
0.10	≥ 0.90	1.000	8.71	0.885
0.05	≥ 0.95	1.000	9.09	0.902
0.02	≥ 0.98	1.000	9.44	0.926
0.01	≥ 0.99	1.000	9.64	0.943

B. Class Imbalance Robustness

To test whether CP coverage degrades for rare faults, ball fault samples (B007/014/021) are reduced to 10% (~ 57 calibration samples vs. 570 for Normal). Table III compares marginal and class-conditional CP at $\alpha=0.10$.

TABLE III
PER-CLASS COVERAGE UNDER IMBALANCE.

Class	Samples	Marginal	Class-Cond.
Normal	3793	1.000	1.000
IR007/014/021	949–959	0.996–1.000	0.996–1.000
OR007/014/021	952–956	1.000	1.000
B007 [†]	948	0.990	0.984
B014 [†]	953	1.000	1.000
B021 [†]	953	1.000	1.000

[†]Rare (10% of normal). All others ≥ 0.996 .

Marginal CP is inherently robust to this imbalance: all classes achieve $>98.9\%$ coverage, including the downsampled ball faults, because APS pools calibration information across all classes into a single threshold $\hat{q}$. Class-conditional CP slightly *decreases* coverage for B007 ($99.0\% \rightarrow 98.4\%$): per-class quantiles suffer high variance from only ~ 57 samples ($1/(n_y+1) \approx 1.7\%$ vs. the marginal $1/(n+1) \approx 0.01\%$). The implication is direct: **practitioners do not need class-conditional calibration** even under severe imbalance, provided the total calibration set yields a stable marginal quantile (Fig. 3).

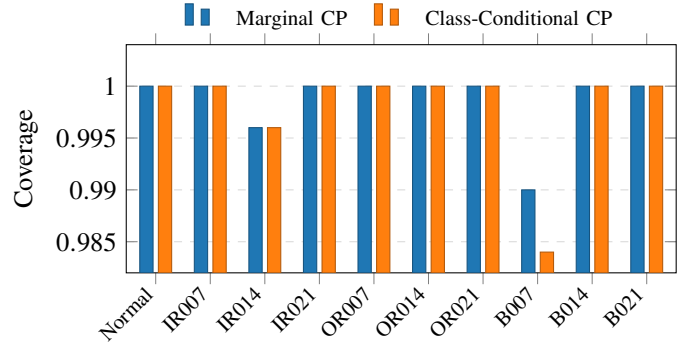


Fig. 3. Per-class coverage under imbalanced training and domain shift.

C. Robustness to Domain Shift Severity

Table IV evaluates CP under three shift levels. Even under severe shift ($99.9\% \rightarrow 86.3\%$ accuracy), CP maintains $\geq 99.9\%$ coverage by widening sets from 6.54 to 8.71: as the confidence distribution flattens, the calibration quantile rises, admitting more classes to preserve the guarantee. Importance-weighted CP [7] gave no efficiency gain over APS, likely because the required likelihood ratios are poorly estimated from high-dimensional features at these sample sizes.

TABLE IV
CP ROBUSTNESS ACROSS SHIFT SEVERITY ($\alpha=0.10$, APS).

Scenario	Acc.	Coverage	Avg. Size	AutoRate
In-Domain (0–1)	0.999	1.000	6.54	0.105
Mild (0–1 $\rightarrow$ 2)	0.996	1.000	7.72	0.082
Severe (0–1 $\rightarrow$ 3)	0.863	1.000	8.71	0.045

D. Selective Prediction for Human-AI Collaboration

Table V compares THR and APS for selective prediction [18]. THR loses all selectivity under shift: 100% singletons at 86.3% accuracy due to softmax concentration. APS identifies 4.5% of samples as automatically classifiable with **100% accuracy** at perfect coverage. Across the full α -curve (Fig. 4), APS holds $\geq 99.7\%$ automated accuracy up to $\alpha = 0.28$ (12.2% automation), degrading gracefully as coverage is

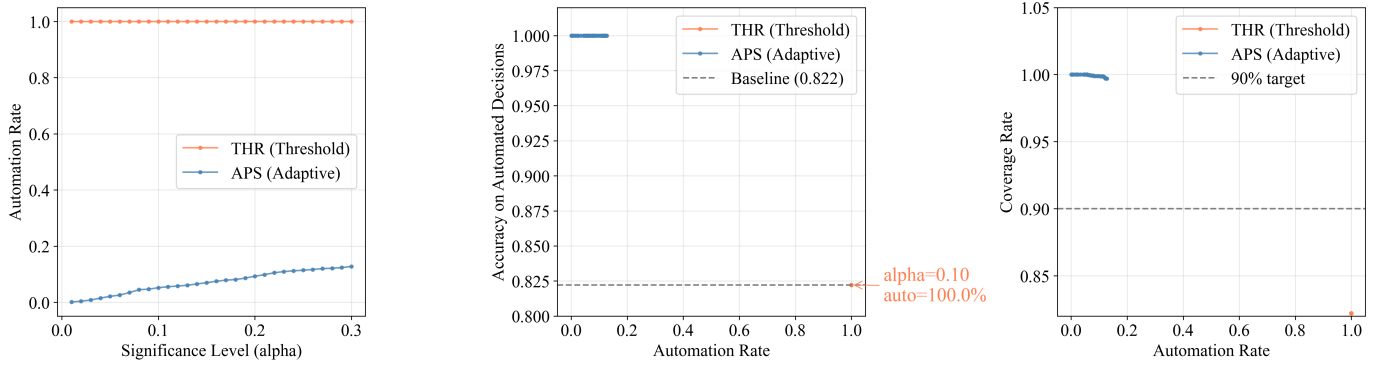


Fig. 4. Selective prediction under domain shift.

relaxed. This contrast is instructive: THR appears maximally automated yet its singletons are no more reliable than the raw classifier, whereas APS automates a smaller fraction but with a coverage guarantee attached to every accepted prediction. Automation rate alone is thus a misleading objective; what matters operationally is the accuracy of the automated subset, which only rank-based scoring keeps near unity under shift.

TABLE V
SELECTIVE PREDICTION UNDER DOMAIN SHIFT ($\alpha=0.10$).

Score	AutoRate	AutoAcc	Coverage
THR	1.000	0.863	0.863
APS	0.045	1.000	1.000

E. Score Function Comparison

Table VI compares all methods at $\alpha = 0.10$ under severe shift. THR fails with only 86.3% coverage—exactly the raw accuracy—because its absolute softmax scores stay concentrated under shift [3]. MC Dropout reaches only 88.5% at $T = 50$ passes, confirming that weight-space epistemic uncertainty does not substitute for shift-aware calibration. APS and RAPS use rank-based scoring, invariant to monotone transformations of the softmax vector and thus robust to the inflated confidence that breaks THR. APS achieves 100% coverage with large sets (8.71 on average); RAPS compresses these to 3.82—a 56% reduction—by penalizing deep ranked tails, still achieving 95.6% coverage.

TABLE VI
SCORE FUNCTION COMPARISON ($\alpha=0.10$, SEVERE SHIFT).

Method	Coverage	Avg. Size	AutoRate
THR	0.863	1.00	1.000
APS	1.000	8.71	0.045
RAPS	0.956	3.82	0.045
MC Dropout	0.885	1.18	—

The dichotomy between absolute and rank-based scoring underlies these results. Absolute scores (THR, MC Dropout) inherit the classifier’s miscalibration as shift intensifies, whereas

rank-based APS and RAPS depend only on probability ordering, which stays informative even when softmax values inflate—hence the gap between the two families widens with shift severity. RAPS recovers most of APS’s robustness at a fraction of the set size, offering the best tradeoff for human-in-the-loop diagnosis, while APS remains preferable for safety-critical deployments. This ordering is consistent across all three α levels tested, confirming that the rank-based advantage is not an artifact of a specific error tolerance (Fig. 5).

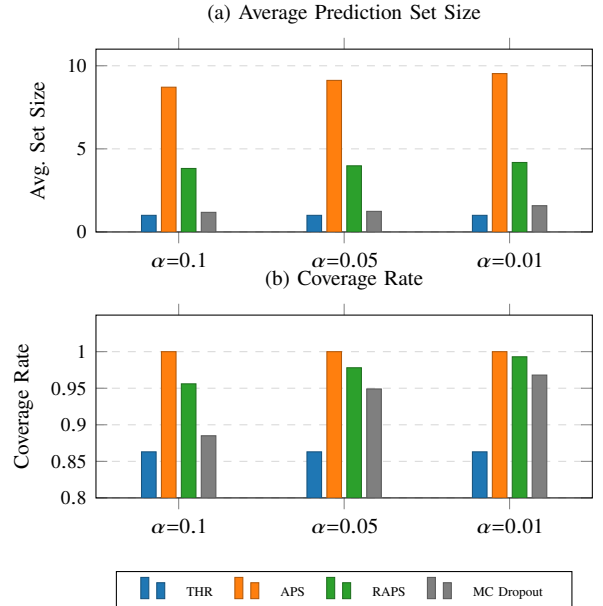


Fig. 5. (a) Average prediction set size of THR, APS, RAPS, and MC Dropout at three α levels; (b) coverage rate of the same methods at three α levels.

F. Deployment Guidelines

Table VII summarizes recommended configurations. The choice of α trades off automation and coverage, from full human review (APS, $\alpha=0.05$) to partial automation (CSD+APS, $\alpha=0.28$, 12.2% auto at 99.7% accuracy). These are not mutually exclusive: a deployment can run multiple α levels in parallel through increasingly conservative thresholds.

TABLE VII
DEPLOYMENT GUIDELINES UNDER DOMAIN SHIFT.

Scenario	Method	α	Expected Outcome
Safety-critical	APS	0.05	100% cov., large sets
Balanced	RAPS	0.10	95.6% cov., 3.82 avg size
Max automation	CSD+APS	0.28	12.2% auto, 99.7% acc

VI. DISCUSSION

A key insight: CP coverage is robust precisely when the classifier is most uncertain, enlarging prediction sets to compensate for accuracy loss—a property absent from heuristic UQ. MC Dropout coverage instead degrades with accuracy, as dropout variance captures epistemic uncertainty about weights, not covariate shift.

Why THR fails: THR exposes a fundamental pitfall in applying CP to deep classifiers. Its scores $1 - \hat{\pi}_y(x)$ cluster near zero during in-domain calibration, yielding a near-zero quantile $\hat{q}$; under shift, overconfident softmax outputs [3] keep nearly every test score below $\hat{q}$, producing singletons with no coverage protection. APS and RAPS avoid this by accumulating probability in rank order, which stays sensitive to uncertainty even when the top-class value is inflated.

When does marginal CP suffice: That marginal calibration matches or exceeds class-conditional CP under imbalance reflects a bias–variance tradeoff. Class-conditional CP removes the cross-class coverage gap but estimates each $\hat{q}_y$ from only n_y samples, a finite-sample penalty of order $1/(n_y+1)$; when rare classes retain tens of points ($n_y \approx 57$ here), this variance dominates and marginal pooling wins, making marginal CP the default for data-scarce settings.

Threats to validity: Three caveats bound these conclusions. First, shift is induced solely through motor load; speed, sensor, or geometry shifts may violate exchangeability more severely. Second, imbalance is simulated by subsampling. Third, all experiments use CWRU ($K=10$); finer taxonomies could enlarge sets.

VII. CONCLUSION

This paper presented the first systematic evaluation of conformal prediction for bearing fault diagnosis under domain shift, class imbalance, and human-AI collaboration. Three findings follow. First, CSD converts classifier uncertainty into an actionable protocol, enabling 4.5% full automation at 100% accuracy. Second, score-function choice is decisive: THR fails under shift while rank-based APS and RAPS hold valid coverage, RAPS giving the best tradeoff (95.6% coverage, 3.82 avg. set size). Third, CP reaches $\geq 99.9\%$ coverage where MC Dropout falls to 88.5%, and marginal CP suffices under imbalance without class-conditional calibration.

Beyond these results, CP reframes uncertainty quantification around a formal guarantee rather than a heuristic score: as a post-hoc wrapper on any pretrained classifier, it adds negligible overhead and needs no retraining, staying compatible with legacy edge-deployed models.

Future work: pairing CP with domain adaptation to shrink APS sets under shift, plus Online CP and cost-sensitive deferral by fault criticality.

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